

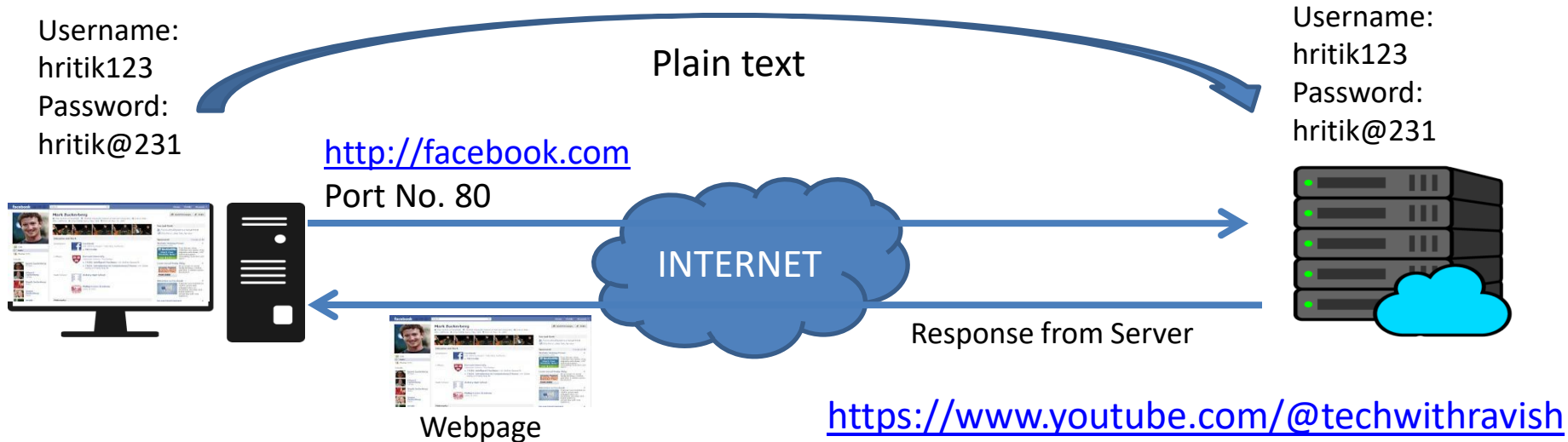
Application Layer

The Application Layer is the topmost layer in the OSI (Open Systems Interconnection) model and is responsible for providing network services directly to end-users or applications. It serves as the interface between the user's software and the network infrastructure.



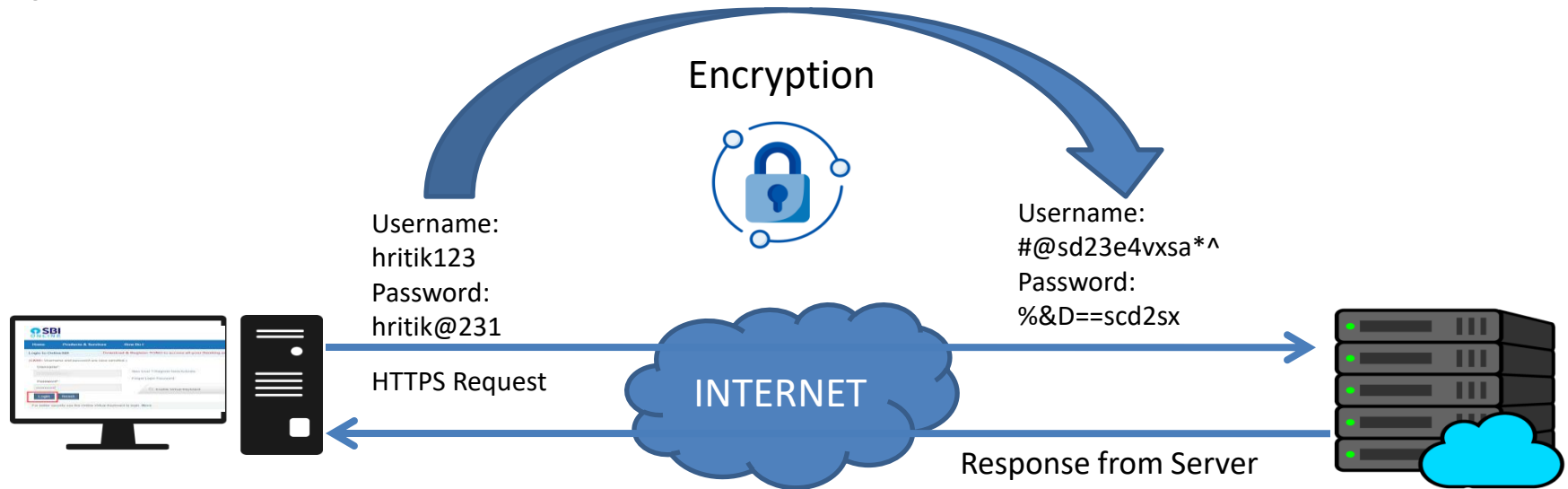
Some commonly used protocols operate at the Application Layer:

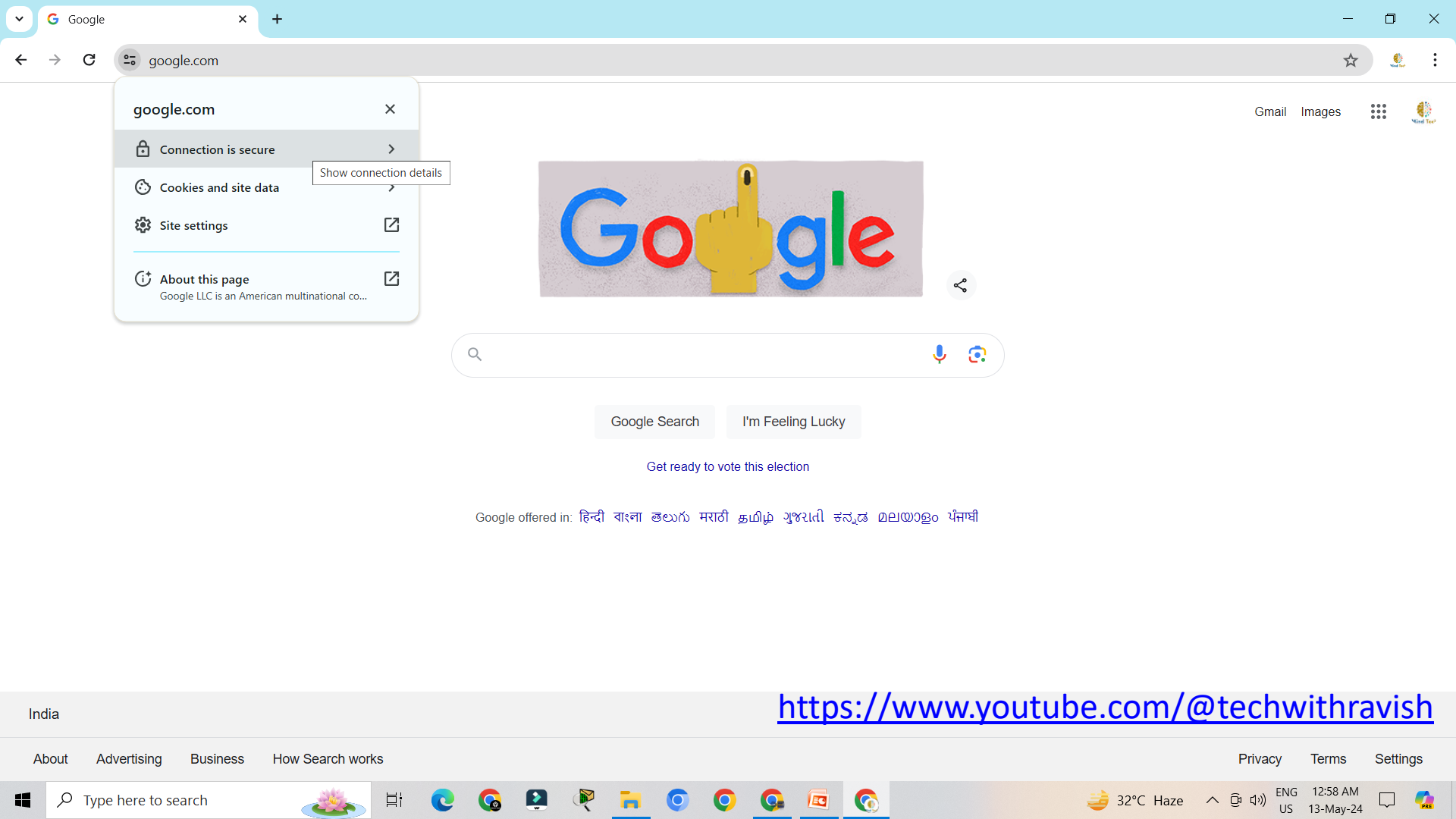
HTTP (Hypertext Transfer Protocol): Used for transferring hypertext requests and information between web browsers and web servers. It forms the basis of communication for the World Wide Web.



HTTPS (Hypertext Transfer Protocol Secure):

A secure version of HTTP that incorporates encryption and authentication mechanisms provided by SSL/TLS protocols for secure communication over the internet.





FTP (File Transfer Protocol): Enables the transfer of files between a client and a server on a network. It allows users to upload, download, list, rename, and delete files on remote servers.

logging in, navigating directories, transferring files, and terminating the session

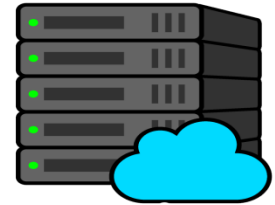
CONTROL CHANNEL (TCP Port 21)



FTP Client



DATA CHANNEL (TCP Port 20)



FTP Server

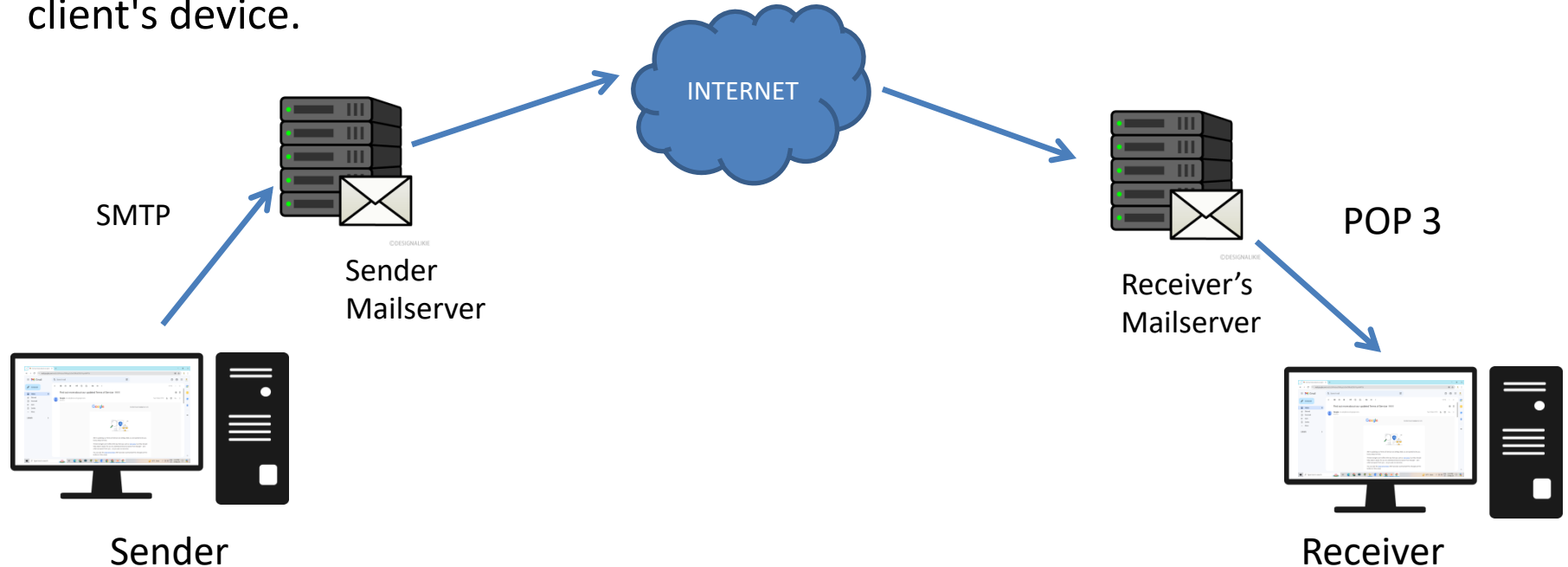


Upload/Download files

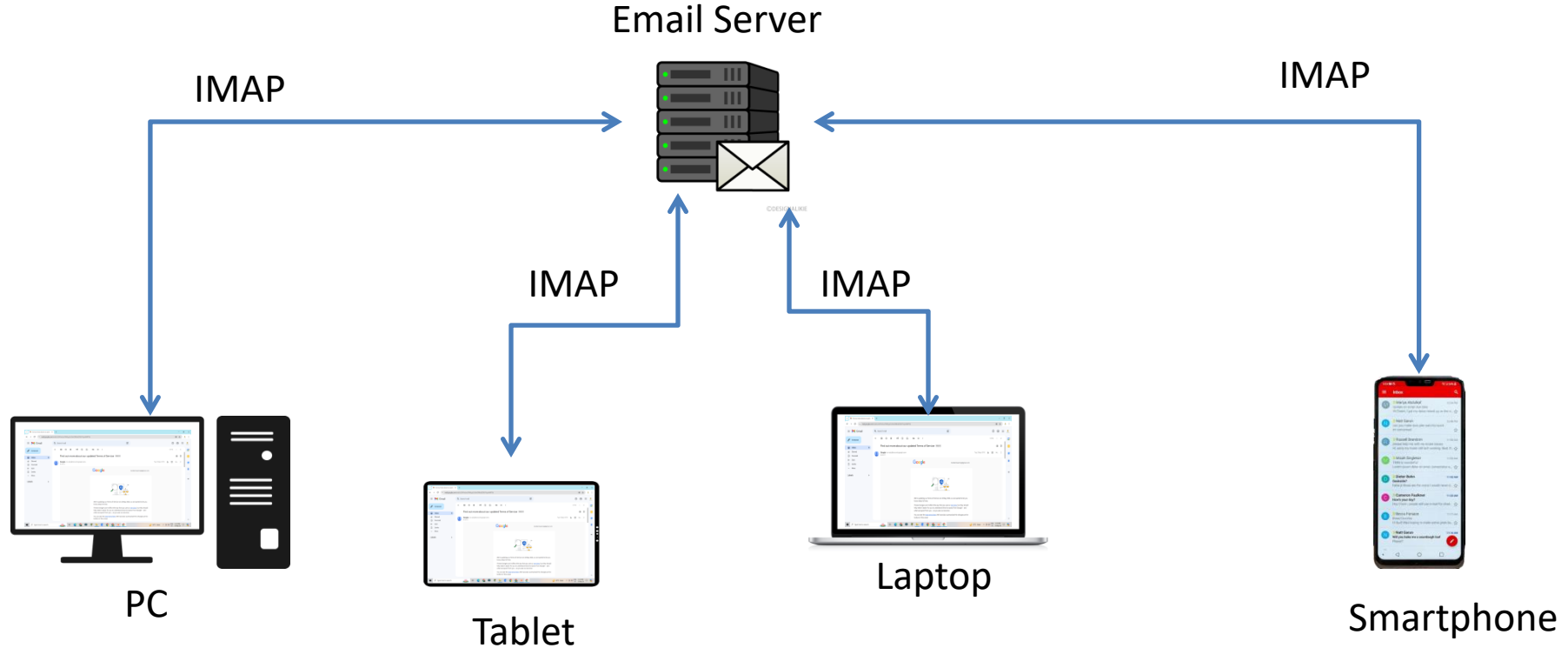
<https://www.youtube.com/@techwithravish>

SMTP (Simple Mail Transfer Protocol): Used for sending email messages between email servers. It defines how email messages are transferred and routed across networks.

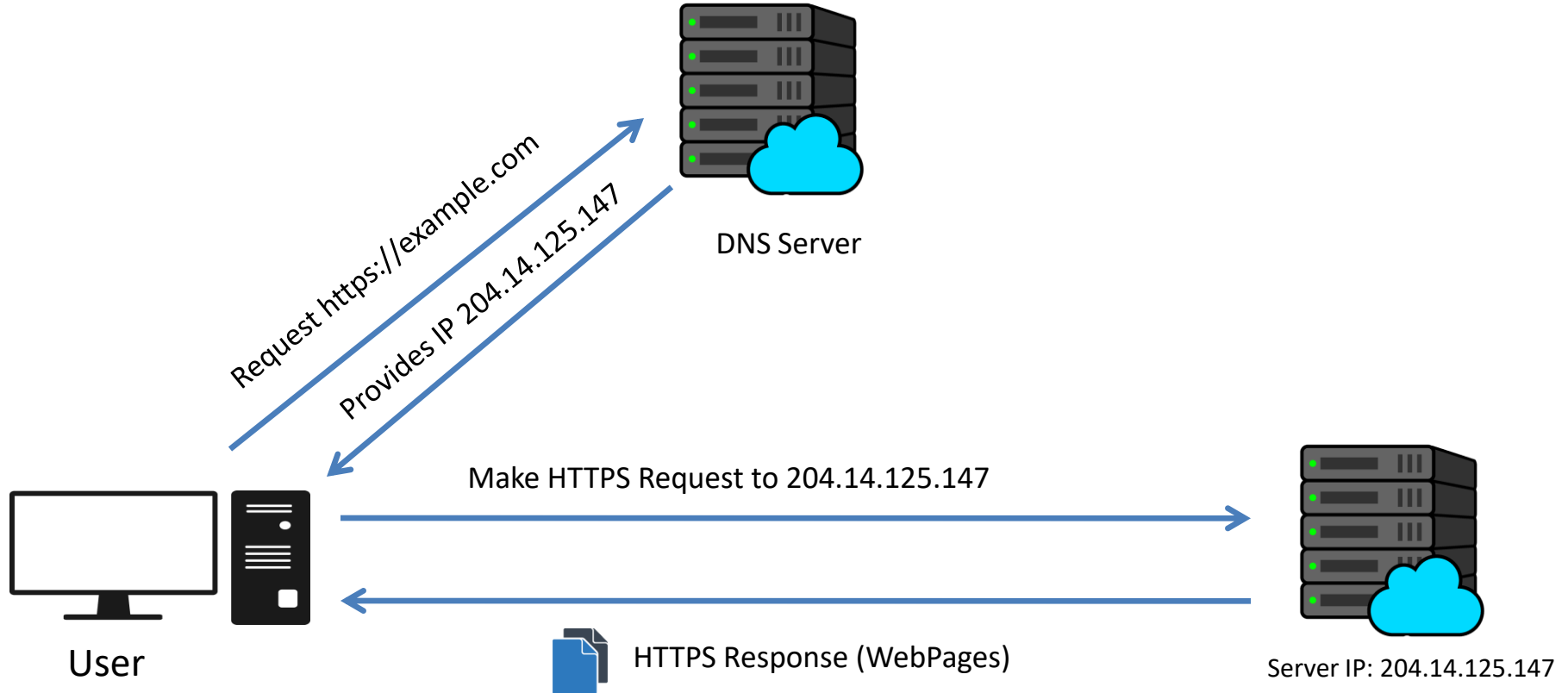
POP3 (Post Office Protocol version 3): Allows email clients to retrieve email messages from a remote server. It is commonly used for downloading emails to the client's device.



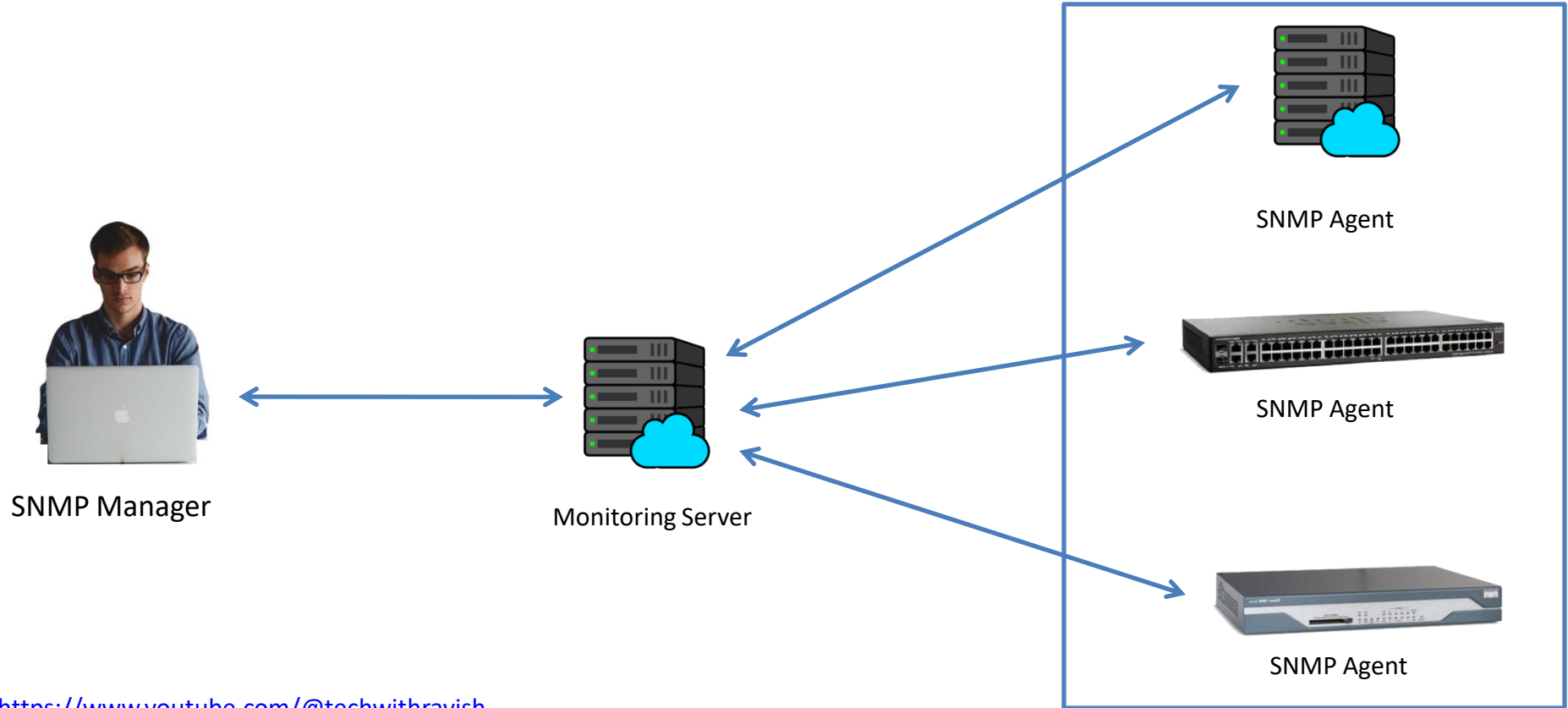
IMAP (Internet Message Access Protocol): Similar to POP3, IMAP allows email clients to access and manage email messages stored on a remote server. It provides more advanced features for email management compared to POP3.



DNS (Domain Name System): Resolves domain names to IP addresses and vice versa. It translates human-readable domain names (e.g., www.example.com) into numerical IP addresses used by computers to communicate over a network.



SNMP (Simple Network Management Protocol): Used for managing and monitoring network devices such as routers, switches, and servers. It allows network administrators to collect information and manage network devices remotely.



Telnet: Allows remote access to servers or devices over a network. It provides a command-line interface to interact with remote systems as if you were physically present at the device.

